Respiration- Aerobic and Anaerobic respiration

This topic will cover the differences between aerobic and anaerobic.

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Title - Aerobic and Anaerobic Respiration

What is the word and symbol equation for both aerobic and anaerobic respiration?

Can you balance both?

(Hint – think about what we take in that plants give out)

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DO NOW

Aerobic Respiration Word Equation

Glucose + Oxygen → Water + Carbon Dioxide + (Energy)

Symbol Equation

$$C_6H_{12}O_6 + 6O_2 \rightarrow 6H_2O + 6CO_2 + (ATP)$$

Anaerobic Respiration

Glucose → Lactic Acid + (Energy)

Symbol Equation

$$C_6H_{12}O_6 \rightarrow 2C_3H_6O_3 + (ATP)$$

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HOME LEARNING

Monday, 21 April 2025

Aerobic and Anaerobic Respiration

Learning objectives:
By the end of the lesson you will be able to:

- Describe cellular respiration as an exothermic reaction which is continuously occurring in living cells.
- Compare the processes of aerobic and anaerobic respiration
- Apply both aerobic and anaerobic respiration in the content of exercise.
- Understand the role of glycogen in exercise

Key words:
Energy
Respiration
Glucose.
Aerobic
Anaerobic
Fermentation
Oxygen Debt

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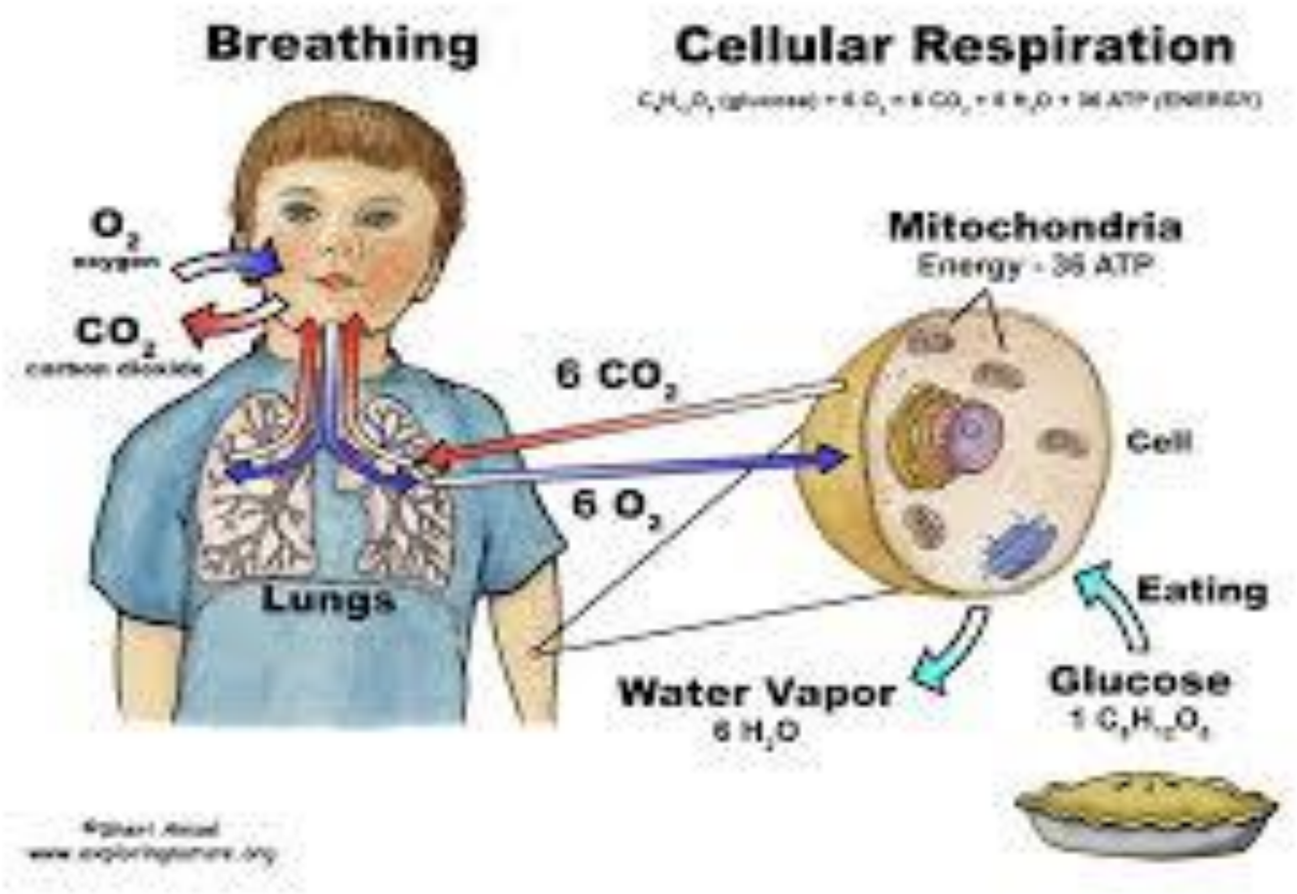
CONNECTION

What is Respiration?

The movement of oxygen from the outside environment to the cells within tissues, and the transport of carbon dioxide in the opposite direction.



Respiration is not the same as breathing. IT does not just take place in the lungs.

It takes place in EVERY CELL of your body in the mitochondria.

Muscles cells contain

What is the difference between ‘breathing’, ‘respiration’ and ‘gaseous exchange’?

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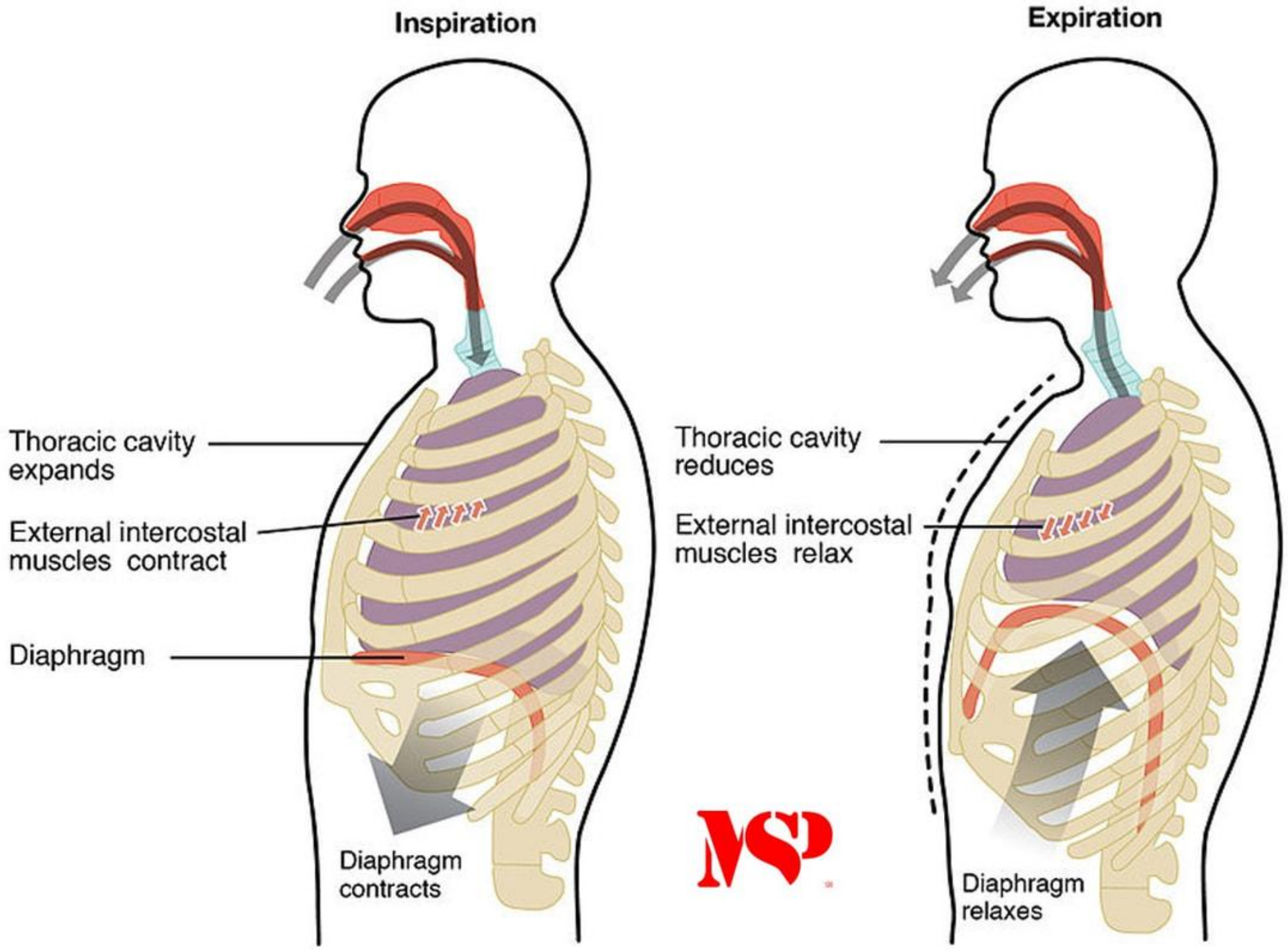
ACTIVATION



What is the difference between 'breathing', 'respiration' and 'gaseous exchange'?

- Breathing is the mechanical movement of air going in and out of our lungs.
- Respiration is the chemical process where oxygen and glucose are converted into energy. Takes place in the mitochondria.
- Gaseous Exchange is the delivery of oxygen from lungs to the bloodstream and carbon dioxide from the bloodstream into the lungs.

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Response to exercise

- During exercise the human body reacts to the increased demand for energy.
- The heart rate, breathing rate and breath volume increase during exercise to supply the muscles with more oxygenated blood.

- This happens because they need lots of oxygen to get rid of the lactic acid that has built up in their body.
- If the demand of oxygen cannot be met the body then falls into anaerobic respiration

Not enough oxygen!

lactic acid + oxygen → carbon dioxide + water

When there is not enough oxygen in the body we call it **<u>Oxygen Debt</u>**

The lactic acid produced in anaerobic respiration soaks the muscle cells causing fatigue and cramps.

Add in the Symbol equation (Lactic acid is half of glucose.)

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Symbol Equation

$$C_3H_6O_3 + 3O_2 \;\square\; 3CO_2 + 3H_2O$$

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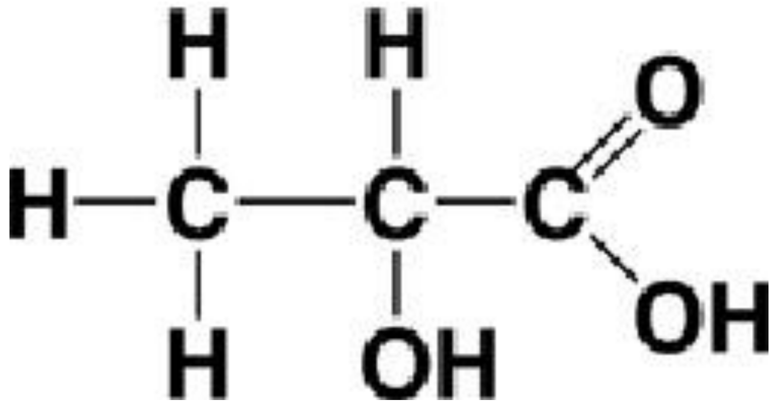
ACTIVATION

Oxygen debt and the Liver

- Blood flowing through the muscles transports the lactic acid to the liver where it is converted back into glucose.
- During long periods of vigorous activity:
- lactic acid levels build up
- glycogen reserves in the muscles become low as more glucose is used for respiration, and additional glucose is transported from the liver.



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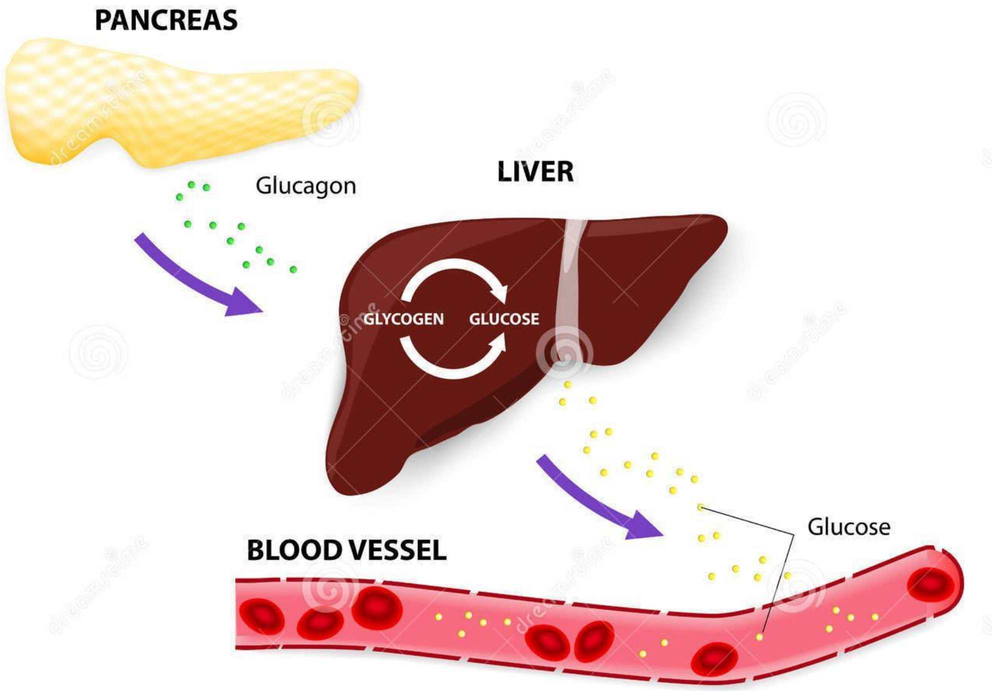
ACTIVATION



Lactic acid is taken to the liver by the blood, and either:

1. Oxidised to carbon dioxide and water, or

2. converted to glucose, then glycogen - glycogen levels in the liver and muscles can then be restored

Glucose that is not used in respiration can be stored as **glycogen** in the liver and in muscle cells.

This can then be converted back to glucose for use during exercise.

Anaerobic respiration in plants

Plants, just like animals, respire anaerobically when oxygen is in low supply.

- **In animals, lactic acid is produced**
- **In plants, ethanol and carbon dioxide are produced**

Glucose ethanol + carbon dioxide

$$C_6H_{12}O_6 \rightarrow 2C_2H_5OH + 2\ CO_2$$

The type of anaerobic respiration that produces ethanol and carbon dioxide is called **fermentation**

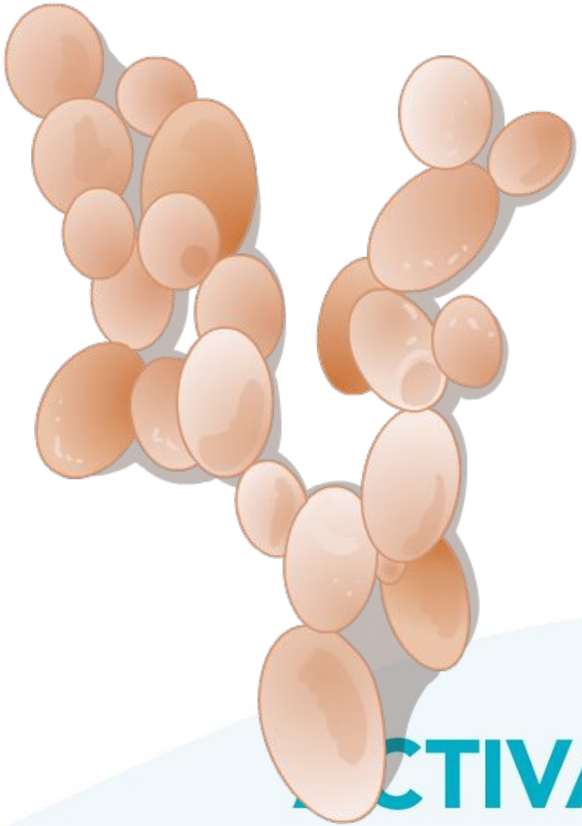
Anaerobic respiration in plants

This can occur in the roots when a plant is growing in boggy or waterlogged soil

Microbes are microorganisms not visible to the naked eye

There are three main types of microbes:

bacteria **viruses** **fungi**



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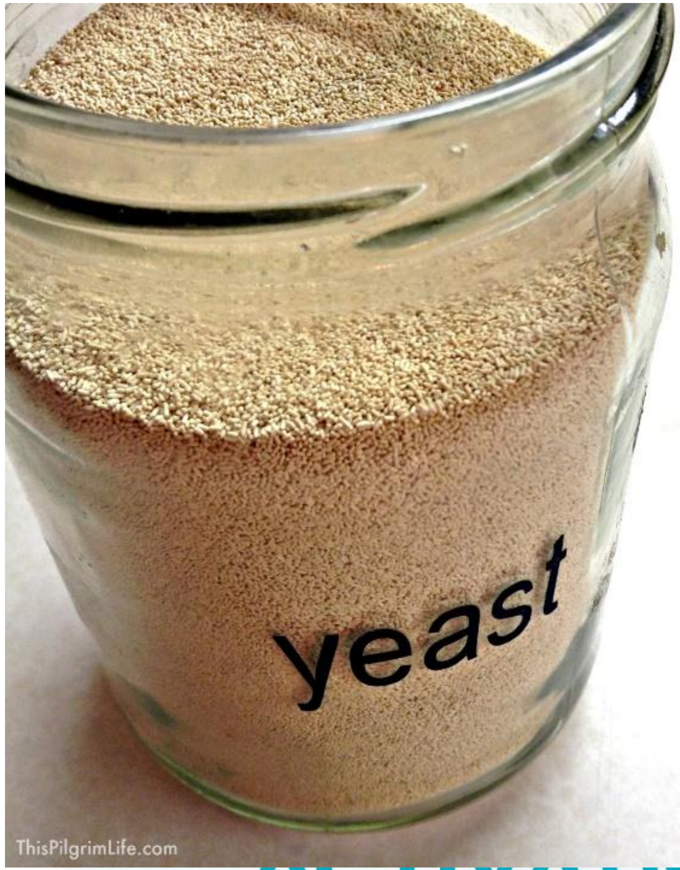
CTIVATION

Yeast

- Microorganisms are tiny living things
- Yeast is a fungus and uses anaerobic respiration
- Some microbes are adapted to survive only in anaerobic conditions, for example, bacteria that live far below the ocean's surface.



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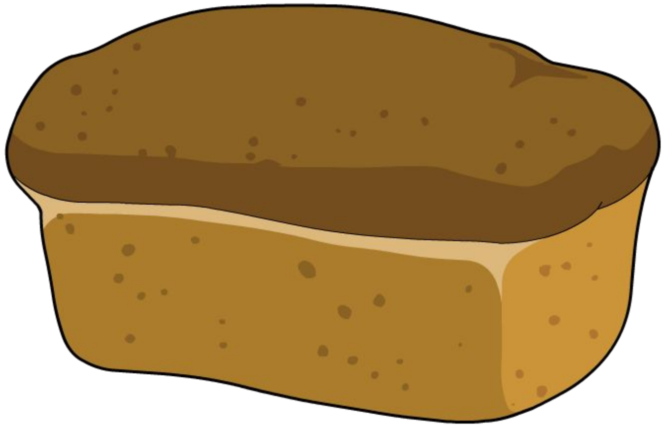
Using yeast

Yeast is a type of fungus and carries out respiration.
The respiration of this microbe can be used in different ways in baking bread and in brewing.



The **aerobic respiration** of yeast is used to make bread rise.

Yeast uses the sugar in bread dough to carry out aerobic respiration:

glucose + oxygen carbon dioxide + water (+ energy)

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ACTIVATION

Using yeast

The **anaerobic respiration** of yeast is used to make beer and wine.

In this case, the yeast respires without oxygen and produces alcohol (ethanol). This process is known as **fermentation**.

Yeast converts the sugar into alcohol by anaerobic respiration:

glucose □ carbon dioxide + ethanol (+energy)

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ACTIVATION

Write down the sentence and sort.

	Aerobic	Anaerobic	Both
glucose is used			
lactic acid is produced			
ethanol is produced			
energy is released			
animals can respire like this			
plants can respire like this			
yeast can respire like this			
sprinting relies mainly on this			
steady jogging relies mainly on this			
during a football match this takes place			
takes place in mitochondria			
leads to oxygen debt			

1 Aerobic or anaerobic? >

Sort each statement below into either ‘aerobic respiration’, ‘anaerobic respiration’ or ‘both aerobic and anaerobic’.

Tick **one** box in each row

		Aerobic	Anaerobic	Both
1.	• glucose is used			✓
2.	• lactic acid is produced		✓	
3.	• ethanol is produced		✓	
4.	• energy is released			✓
5.	• animals can respire like this			✓
6.	• plants can respire like this			✓
7.	• yeast can respire like this			✓
8.	• sprinting relies mainly on this		✓	
9.	• steady jogging relies mainly on this	✓		
10.	• during a football match this takes place			✓
11.	• takes place in mitochondria	✓		
12.	• leads to oxygen debt		✓	

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CONSOLIDATION

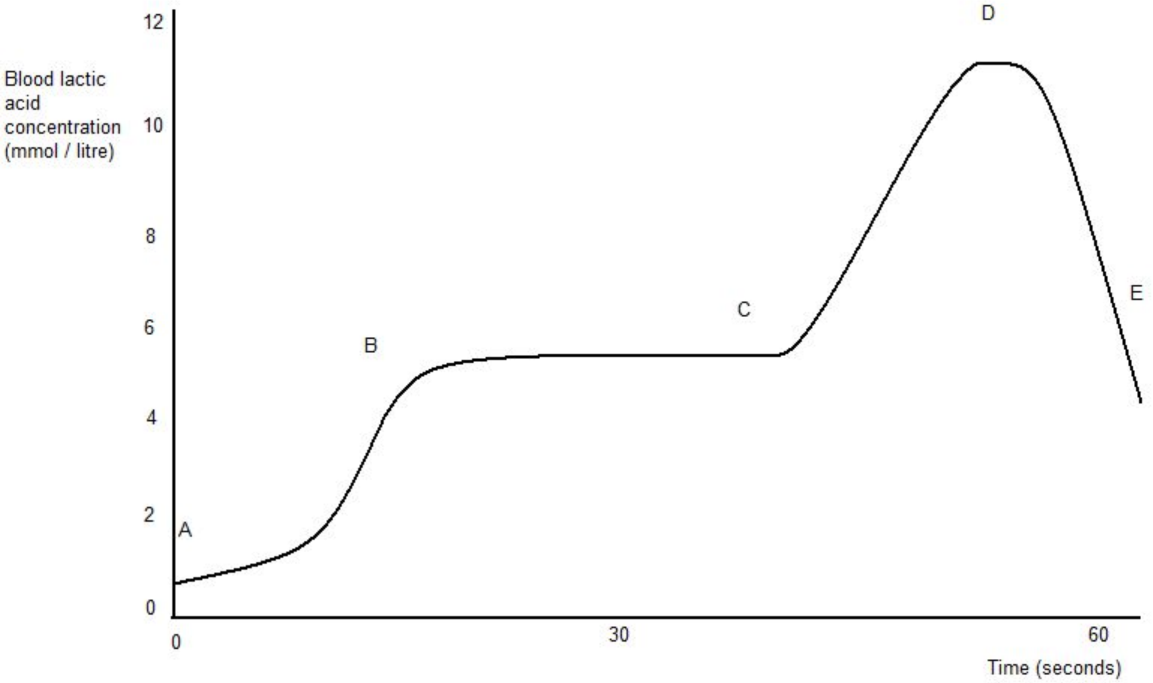
Plenary - Exam practice

- In your exam you will be expected to analyse and evaluate graphs demonstrating aerobic and anaerobic respiration.
- Task- complete the worksheet based on anaerobic respiration submit your answers to SMH

Respiration in athletics

The blood of an athlete was tested before, during and after a 400m race:

TASK 4 – explaining muscle fatigue and oxygen debt



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Complete the following table about each section of the race:
Mark yourself out of 10.

Section of race	Strategy	Anaerobic or aerobic respiration?	Explanation
A to B	Sprint start	Anaerobic (1)	Sudden increase in exercise, not enough oxygen, respires anaerobically, produces lactic acid, and causes increase in lactic acid levels. (1)
B to C	Stops accelerating and maintains pace	Aerobic (1)	Breathing rate will have increased, oxygen levels also increased, can now respire aerobically, no more lactic acid produced. Level stays same as no extra oxygen available to break it down. (2)
C to D	Sprint finish	Anaerobic (1)	Sudden increase in exercise, not enough oxygen, respires anaerobically, produces lactic acid, and causes increase in lactic acid levels. (1)
D to E (after the race has	Warm down, then rest	Aerobic (1)	Decrease in intensity of exercise, breathing rate will still be high, extra oxygen can break down lactic acid. Paying back oxygen debt, decreasing amount of lactic acid as

Homework

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